

Solution – Exercise II

HiveQL, Create and work with External Tables on IMDb Data

Solution

Prerequisites:

- Setup Google Cloud SDK
- Start VM instance
- Pull docker container `marcelmittelstaedt/hive_base:latest`
- Start docker container: `docker run -dit --name hive_base_container -p 8088:8088 -p 9870:9870 -p 9864:9864 marcelmittelstaedt/hive_base:latest`
- Get into docker container
- Start Hadoop and Hive Shell:
  - `start-all.sh`
  - `hive`

Solution

Exercise 1-4:

1. Download and unzip https://datasets.imdbws.com/name.basics.tsv.gz

```
wget https://datasets.imdbws.com/name.basics.tsv.gz
gunzip name.basics.tsv.gz
```

2. Create HDFS directory **/user/hadoop/imdb/name_basics/** for file name.basics.tsv

```
hadoop fs -mkdir /user/hadoop/imdb/name_basics
```

3. Put TSV file to HDFS:

```
hadoop fs -put name.basics.tsv /user/hadoop/imdb/name_basics/name.basics.tsv
```

Solution

Exercise 1-4:

4. Create Hive Table `name_basics`:

```
hive > CREATE EXTERNAL TABLE IF NOT EXISTS name_basics(
    nconst STRING,
    primary_name STRING,
    birth_year INT,
    death_year STRING,
    primary_profession STRING,
    known_for_titles STRING
    ) COMMENT 'IMDb Actors' ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t' ST
ORED AS TEXTFILE LOCATION '/user/hadoop/imdb/name_basics'
TBLPROPERTIES ('skip.header.line.count'='1');
```

Solution

Exercise 5:

a) How many movies and how many TV series are within the IMDB dataset?

```
hive >  SELECT m.title_type, count(*)
        FROM title_basics m GROUP BY m.title_type;

tvMovie 156517
movie 759083
tvEpisode 9939190
tvSeries 306648
[…]

Time taken: 32.908 seconds, Fetched: 11 row(s)
```

b) Who is the youngest actor/writer/… within the dataset?

```
hive >  SELECT * FROM name_basics n
        WHERE n.birth_year = ( SELECT MAX(birth_year) FROM name_basics);
```

Solution

Exercise 5:

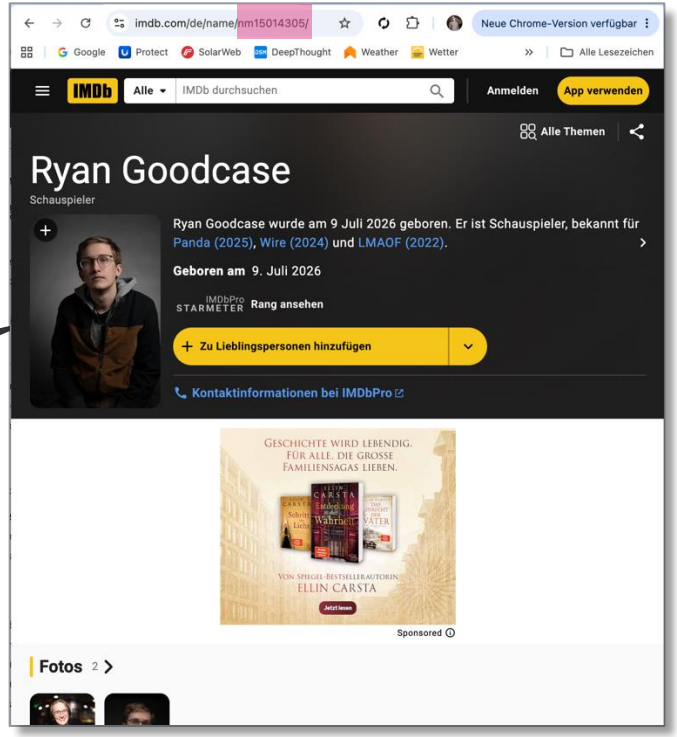
b) Who is the youngest actor/writer/… within the dataset?

```
hive >  SELECT * FROM name_basics n
        WHERE n.birth_year = ( SELECT MAX(birth_year)
                               FROM name_basics);
```

And it's Ryan Goodcase.

```
nm15014305  Ryan Goodcase   2026    NULL    actor   tt42913925,tt33066066,tt21871416,tt0890935

Time taken: 65.166 seconds, Fetched: 1 row(s)
```



Solution

Exercise 5:

c) Create a list (`m.tconst, m.original_title, m.start_year, r.average_rating, r.num_votes`) of movies which are:
- equal or newer than year 2010
- have an average rating equal or better than 8,1
- have been voted more than 100.000 times

```
hive >   SELECT m.tconst, m.original_title, m.start_year, r.average_rating, r.num_votes
         FROM title_basics m JOIN title_ratings r on (m.tconst = r.tconst)
         WHERE r.average_rating >= 8.1 and m.start_year >= 2010 and m.title_type = 'movie'
         and r.num_votes > 100000
         ORDER BY r.average_rating desc, r.num_votes DESC;

tt1375666   Inception   2010    8.8 2877777
tt28089784  Satluj  2026    8.8 106836
tt0816692   Interstellar    2014    8.7 2618469
tt23849204  12th Fail   2023    8.7 178493
tt15097216  Jai Bhim    2021    8.6 236831
tt10189514  Soorarai Pottru 2020    8.6 133175
tt1853728   Django Unchained    2012    8.5 1919071
tt6751668   Gisaengchung    2019    8.5 1201335
tt2582802   Whiplash    2014    8.5 1186147
tt1675434   Intouchables    2011    8.5 1044926
[…]
```

Solution

Exercise 5:

d) How many movies are in list of c)?

```
hive >    SELECT count(*)
          FROM title_basics m JOIN title_ratings r on (m.tconst = r.tconst)
          WHERE r.average_rating >= 8.1 and m.start_year >= 2010 and m.title_type = 'movie'
          and r.num_votes > 100000;

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```

Solution

Exercise 5:

e) We want to know which years have been great for cinema.
Create a list with one row per year and a related count of movies which:
- have an average rating better than 8
- have been voted more than 100.000 times
ordered descending by count of movies.

```
hive >  SELECT m.start_year, count(*)
        FROM title_basics m JOIN title_ratings r on (m.tconst = r.tconst)
        WHERE r.average_rating > 8 AND m.title_type = 'movie'
        AND r.num_votes > 100000
        GROUP BY m.start_year
        ORDER BY count(*) DESC;

1995    8
2004    7
1957    6
2019    6
2018    6
[…]
```




Solution

Exercise 5:

So 1995 seems to be a really good year for cinema, 8 really good movies have been releases, but which are they?

```
hive > SELECT
            m.tconst, m.original_title, m.start_year, r.average_rating,
            r.num_votes
       FROM title_basics m JOIN title_ratings r ON (m.tconst = r.tconst)
       WHERE
            r.average_rating > 8 AND m.title_type = 'movie'
            AND r.num_votes > 100000 AND m.start_year = 1995
       ORDER BY r.average_rating DESC;

tt0114369   Se7en   1995    8.6 2066783
tt0114814   The Usual Suspects  1995    8.5 1257587
tt0113277   Heat    1995    8.3 820774
tt0114709   Toy Story   1995    8.3 1202483
tt0112573   Braveheart  1995    8.3 1174180
tt0112641   Casino  1995    8.2 629502
tt0112471   Before Sunrise  1995    8.1 385294
tt0113247   La haine    1995    8.1 232594
[…]
```